

Programme	B. Sc. Mathematics Honours			
Course Code	MAT5EJ306			
Course Title	LINEAR PROGRAMMING			
Type of Course	Elective			
Semester	V			
Academic Level	300 - 399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Calculus and Linear Algebra			
Course Summary	Linear Programming is a mathematical modelling technique in which a linear function is maximized or minimized when subjected to various constraints. This technique has been useful for guiding quantitative decisions in business planning, in industrial engineering, and—to a lesser extent—in the social and physical sciences. This course begins with convex sets and extrema of functions for a sound basis of the subject. It then develops into LP problems including Transportation and Assignment problems.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Able to identify and analyse the properties of convex sets, including open and closed sets, convex hulls, and vertices.	An	C	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO2	To demonstrate proficiency in applying optimization techniques such as gradient descent, constrained extrema, and the method of Lagrange multipliers to solve real-valued functions.	Ap	P	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO3	To formulate and solve linear programming problems, including transportation and assignment problems, using techniques such as simplex method and duality.	U	P	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text book	Optimization Methods in Operation Research and System Analysis (4th edition), K.V Mittal, C Mohan, New Age International (P)Limited (2016)	
Module	Unit	Content
I	Module I	
	1	Chapter 1 Section 11: Open and Closed sets in E_n
	2	Section 12: Convex Linear Combination, Convex Sets
	3	Section 13: Intersection of Convex Sets, Convex Hull of a Set Section 14: Vertices or Extreme Points of a Convex Set
	4	Section 15: Convex Polyhedron Section 16: Hyperplanes, Half-spaces and Polytopes
	5	Section 17: Separating and Supporting Hyperplanes (Proof of Theorem 18 is optional) Section 18: Vertices of a Closed Bounded Convex Set (Proof of Theorem 21,22,23 are optional) Section 19: Summary Section 20: Quadratic Forms
II	Module II	
	6	Chapter 2 Section 11: Convex Functions
	7	Section 12: General Problem of Mathematical Programming
	8	Chapter 3 Section 1: Introduction Section 2: LP in Two-Dimensional Space
	9	Section 3: General L P Problem Section 4: Feasible Solutions (Proof of Theorem 1 is optional) Section 5: Basic Solutions Section 6: Basic Feasible Solutions (Proof of Theorem 2,3 are optional) Section 7: Optimal Solution (Proof of Theorem 4,5 are optional) Section 8: Summary
	10	Section 9: Simplex Method Section 10: Canonical Form of Equations Section 11: Simplex Method (Numerical Example) Section 12: Simplex Tableau
	11	Section 13: Finding the First b.f.s; Artificial Variables Section 14: Degeneracy
	12	Section 15: Simplex Multipliers
III	Module III	
	13	Chapter 3 Section 17: Duality in LP Problems
	14	Section 18: Duality Theorems (Proof of Theorem 7,8,9, 10,11 are optional) Section 19: Applications of Duality
	15	Section 20: Dual Simplex Method Section 21: Summary of Simplex Methods (III Revised Simplex Method is optional)
	16	Section 22: Applications of LP
IV	Module IV	

	17	Chapter 4 Section 1: Introduction Section 2: Transportation Problem Section 3: Transportation Array Section 4: Transportation Matrix Section 5: Triangular Basis (Proof of Theorem 1 is optional) Section 6: Finding a Basic Feasible Solution
	18	Section 7: Testing For Optimality
	19	Section 8: Loop in Transportation Array (Proof of Theorem 2 is optional) Section 9: Changing the Basis
	20	Section 10: Degeneracy Section 11: Unbalanced Problem
	21	Section 14: Assignment Problem (Proof of Theorem 3 is optional)
	22	Section 15: Generalized Transportation Problem Exercise Questions in Assignment Problem
V	Open Ended	
		Linear Programming Using Scipy, Prog Reference 1.
		Dual Simplex Solved Programming Exercises in Python from Vanderbei (Reference 1), Prog Reference 2.
		Linear Programming in Python using IBM CPLEX Community Edition. Prog Reference 3.
		Transportation Problem in Python. Prog Reference 4.
		Linear Programming in Julia. Prog Reference 5. Ch 3 Basics of Julia Programming Language, Ch 5 The Simplex Method.
	. References: 1. G. Hadley : Linear Programming Addison-Wesley Pub Co Reading, Mass (1975) 2. S.S. Rao : Optimization – Theory and Applications (2nd Edn.) Wiley Eastern (P) Ltd. New Delhi. 3. Russel L Ackoff and : Fundamentals of Operation Research Maurice W.Sasieni Wiley Eastern Ltd. New Delhi. (1991) 4. Charles S. Beightler, : Foundations of Optimization D.T. Philips & D.J. Wilde (2nd Edn.) Prentice Hall of India, Delhi (1979)	
	Programming References for Open-Ended section: 1. Linear Programming using Scipy, https://python.quantecon.org/lp_intro.html 2. Vanderbei's book homepage: https://vanderbei.princeton.edu/LPbook/ 3. CPLEX Jupyter Notebook: https://github.com/IBMDecisionOptimization/tutorials/blob/master/jupyter/Linear_Programming.ipynb Installation: http://ibmdecisionoptimization.github.io/docplex-doc/README.md.html	

	4. Solving Transportation Problem using Linear Programming in Python: https://machinelearninggeek.com/solving-transportation-problem-using-linear-programming-in-python/ 5. Changhyun Kwon, Julia Programming for Operations Research 2/e , https://www.softcover.io/read/7b8eb7d0/juliabook2/simplex
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Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	1	2	1	3	0	1
CO 2	3	3	3	2	2	1	3	1	3	0	1
CO 3	2	3	3	2	3	1	3	1	3	0	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓